

Introduction

This report presents a CGPA Calculator developed in C++. The program calculates the GPA for a semester and the overall CGPA based on course grades and credit hours.

Features

1. **Course Input:** User enters the number of courses, course names, grades, and credit hours.
2. **Grade Validation:** Only valid grades (A, B, C, D, F) are accepted.
3. **Credit Hour Validation:** Credit hours must be between 1 and 6.
4. **GPA Calculation:** Semester GPA is calculated based on total grade points and total credits.
5. **CGPA Calculation:** Overall CGPA is calculated by including previous semester data.
6. **Formatted Output:** Results are displayed in a clean, tabular format.

Code Overview

The program uses:

- **Structures:** Course struct holds course details.
- **Functions:**
 - gradeToPoint() — Converts letter grade to grade point.
 - getValidGrade() — Validates grade input.
 - getValidCreditHours() — Validates credit hour input.
 - displayResults() — Displays formatted results.
- **Vectors:** Stores course data dynamically.

Output

```

C:\Users\iqra0\OneDrive\Documents\codealpha\CodeAlpha_C++ task\task1\cgpa_calculator.exe
=====
CodeAlpha CGPA CALCULATOR
=====

How many courses did you take this semester? 3

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ENTER COURSE DETAILS
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Course 1:
Enter course name: Mathematics
Enter grade for course 1 (A/B/C/D/F): A
Enter credit hours for course 1 (e.g., 3): 3

Course 2:
Enter course name: Physics
Enter grade for course 2 (A/B/C/D/F): B
Enter credit hours for course 2 (e.g., 3): 4

Course 3:
Enter course name: English
Enter grade for course 3 (A/B/C/D/F): A
Enter credit hours for course 3 (e.g., 3): 2

=====
                        RESULT SUMMARY
=====
Course      Grade    Credits    Points
-----
Mathematics    A        3.0       12.00
Physics        B        4.0       12.00
English        A        2.0        8.00
-----
Total Credit Hours:          9.0
Total Grade Points:        32.00
-----

Semester GPA : 3.56 / 4.00

Do you want to calculate your overall CGPA? (y/n): y
Enter total credit hours from previous semesters: 30
Enter total grade points from previous semesters: 105

=====
Overall CGPA : 3.51 / 4.00
=====

Thank you for using the CodeAlpha CGPA Calculator!
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Process exited after 60.97 seconds with return value 0

```

Figure 1: CGPA Calculator Output

Conclusion

The CGPA Calculator demonstrates the use of structures, functions, loops, and input validation in C++. It provides students with a quick way to calculate their GPA and CGPA.

References

1. C++ Documentation. (2026). *Standard Library*. Retrieved from <https://en.cppreference.com>
2. Stroustrup, B. (2018). *The C++ Programming Language*. Pearson.